

Some important questions for midterm

EDU (430)

Lecture No.1 to 22 (topics 01 to 110)

1. Define ICT and its characteristics?

○ **Definition of ICT:**

ICT is concerned with the storage, retrieval, manipulation, transmission or reception of digital data.

○ **Information and Communication Technology:**

"ICT is defined, as a diverse set of technological tools and resources used to communicate, and to create, disseminate, store, and manage information" (Mangal, 1997).

○ **Characteristics of ICT:**

- ICT is the integration of Hardware and Software Technology.
- ICT fosters the dissemination of information and knowledge.
- The nature of ICT is digital and virtual.
- ICT is global, anywhere, any time.
- ICT is a key enabler in the creation of network.

2. Define Local Area Network and its Characteristics?

○ **Local Area Network:**

Local area networks are networks with in a single building or campus or up to a few kilometers in size.

○ **Characteristics of LAN:**

- Low delay
- Very few errors
- Higher speed
- LANs are restricted in size (Functions within 10 KMs).
- Transmission time is bounded.
- Simple network management

3. Define Wide Area Network and its Characteristics?

○ **Wide Area Network:**

It is defined as networking in a large geographical area, often a country or continent.

○ **Characteristics of WAN:**

- WAN uses telephone lines or satellite links as a medium.
- LAN may be connected to WAN.
- It covers different cities or countries.
- WANs are not owned by any one organization but are under collective ownership and management.
- Complex network design
- Error rate is large.
- Speed is less compared to LAN

4. What are Advantages of ICT in Education?

○ **Advantages of ICT in Education:**

- Quick access to information
- Easy availability of updated data
- Connecting Geographically dispersed regions
- Catering to the Individual differences
- Wider range of communication media
- Wider learning opportunities for pupils

5. Define Internet?

○ **Internet:**

Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words 'interconnection' and 'network'. Through this, each computer user can communicate and share common resources with other users.

6. Define Web Server?

○ **Web Server:**

It is a programmer and a computer clubbed into one entity which responds to the requests from web browsers for internet resources. The primary function of a web server is to deliver web pages to clients' e. g. Apache HTTP Server, Microsoft Internet Information Services (IIS), Sun Java System Web Server.

7. Define Web Browser?

- **Web Browser:**

It is a program that allows a user to display and interact with a 'hypertext' documents e. g. Internet Explorer, Net Scape Navigator, Mozilla Firefox, Google Chrome.

8. Difference between Website and Web Page?

- **Website:**

Website is defined as a collection of related pages on particular information.

- **Web Page:**

It is a file of information stored on a web server.

9. Define Homepage and search engines?

- **Homepage:**

Home page is the first hypertext document to be shown when a user follows a link to the web server.

- **Search Engines:**

Search engines can be defined as special tool to search for information on the net.

- AltaVista
- Yahoo
- Google
- Bing
- AOL

10. Define Intranet and its Characteristics?

- **Intranet:**

Intranet can be defined as collection of private computer networks within an organization.

- **Characteristics:**

- Internal or private network of an organization based on internet technology (such as hypertext and protocols) and accessed over the internet.
- An intranet is meant for the exclusive use of the organization and its associates (customers, employees, members, suppliers, etc.) and is protected from unauthorized access with security systems such as firewalls.
- An intranet should be closed off to the general public and require a login for authorized users.

11. Difference between Uploads and Downloads?

Both upload and download refer to the process of transferring a file.

○ **Uploading:**

Upload means to transfer a file or files from your own computer to another computer. Uploading is the process of copying files from your computer to a remote location. For example, you might transfer a file from your home PC to the Yahoo! computer that stores your Web Hosting files.

○ **Downloading:**

- Download means to transfer a file from another computer to your own computer.
- Downloading is the process of copying files to your computer from a remote location. For example, you might download a file from your Web Hosting account to your home PC.

12. Define Bookmarks?

○ **Bookmarks:**

- Book marks can be defined as, a collection of direct links to predefined web pages which are stored in your web browser.
- Bookmarks can be created and managed by the user themselves. Most browsers come packaged with several existing bookmarks.

13. Define Learning Aids?

○ **Learning Aids:**

Learning aids can be defined as any device/Aids which help self-learning/group learning.

14. Define Teaching Aids and Advantages of Teaching Aids?

○ **Teaching Aids:**

- Teaching aids are the physical tools used to convey information in the classroom.
- Teaching Aids are any device which can be used to make the communication more effective.
- Teaching Aids are any device which can be used to make the learning experience more concrete, more realistic and more dynamic.
- 50 % of what we HEAR and SEE
- 70 % of what we SAY
- 90 % of what we SAY and DO

○ **Advantages of Teaching Aids:**

- Help to learn more and remember for long.
- Motivate the learner.
- Give clarity to learning.

- Give reality and vividness to learning situations.
- Make abstract concrete
- Reduce verbalism.
- Provide variety in methods of teaching and learning.
- Make learning interesting meaningful and permanent.
- Develop deeper understanding.
- Arouse curiosity and self-activity.
- Saving of energy and time
- Spread of education on a mass scale.

15. What are the Characteristics of Good Teaching Aids?

○ **Characteristics of Good Teaching Aids:**

- They should be meaningful and purposeful
- They should be accurate in every respect
- They should be simple
- They should be cheap
- As far as possible, they should be improvised.
- They should be large enough to be properly seen by the students for whom they are meant.
- They should be up to date
- They should be easily portable
- They should be according to the mental level of the student.
- They should be motivating the learner

16. Define Hyper Text Markup Language and its history?

○ **Hyper Text Markup Language (HTML):**

Hyper Text Markup Language (HTML) is a standard language used for developing web pages'. Hyper Text Markup Language is a markup language that web browsers use to interpret and compose text, images and other material into visual web pages.

○ **History of HTML:**

- In 1980, a markup language, i.e., Standard Generalized Markup Language or SGML was developed.
- HTML was evolved from SGML.
- Presently, HTML is advanced to DHTML and then to XML (Extensible Markup Language) and now to VRML (Virtual Reality Modeling Language).

17. Define Teaching as a Noble Profession?

○ **Teaching as a Noble Profession:**

- Teaching is regarded as the noblest of all professions in the world.
- A good teacher not only shows the right path that the students should follow but also prepares the human resource for the further development of the nation.
- Teaching has the power to create a healthy and peaceful world.
- It has the capacity to inspire lifelong learning and a passion for knowledge, understanding, and innovation.

18. Describe Teaching Machines?

○ **Teaching Machines:**

- These were developed by S.L. Pressy.
- It's a piece of device designed to be operated by an individual student for self-learning.
- The student is presented with a question by some form of display on the machine.
- The student is required to respond either to write answer or put a button to indicate the correct response.
- The student will inform the correctness of the answer.
- An account is kept of the responses made by the student.

19. Define Hardware?

○ **Hardware:**

The concept of hardware includes computer components, the physical and tangible parts of the computer, i.e., electrical, electronic and mechanical parts which comprise a computer.

20. Define Personal Computer?

○ **Personal Computer:**

Personal computer (PC), as the name suggests, is intended for personal use, as opposed to the server, which is used by a larger number of people simultaneously, from different locations, often via terminals. If you do not intend to move your computer frequently from one place to another, and at the same time you want maximal price/performance ratio, then you should use a desktop computer. In comparison to laptops or tablet computers, it is much larger in size, inconvenient to carry/move, consumes more electricity but has a much better price/performance ratio. Also, they are much easier to upgrade.

21. What are the Functions of Teaching?

These are the following functions of teaching:

➤ **Preservative Function of Teaching:**

- The main function of TEACHING is to preserve and protect the old values, customs, beliefs, traditions, etc.
- Teaching must help in preserving the vital elements of our heritage.

➤ **Transitive Function of Teaching:**

Teaching transfers knowledge, ideologies, theories, principles and attributes from one generation to another. Thus, it contributes greatly to the general progress of society.

➤ **Progressive Function of Teaching:**

Teaching provides opportunities to recognize and restructure human experiences and as a result of this, there is development of human civilization and culture.

22. Define Mother Board and Protocol?

○ **Mother Board:**

Motherboard, MBO is computer's basic circuit, to which all computer components are connected, directly or indirectly. Devices are connected to the motherboard through a system bus. System bus connects all devices, ensures data flow and communication between different devices using predefined protocols.

○ **Protocol:**

Protocol describes a manner in which communication between devices is defined. It enables them to address each other and defines how they should look for each other on either system bus or network. Buses can, according to the purpose, be divided into:

1. Serial USB, Firmware, etc.
2. Parallel AGP, PCI, etc.
3. Mixed-Hyper Transport, Infini Band, PCI, etc.

23. Define Cache?

○ **Cache:**

Cache is a small capacity memory which allows quick access to data. By storing data from working memory in cache, the speed of communication between processor and RAM is increased. Microprocessors use three levels of fast cache, L1, L2 and L3, used to store often used data.

24. What is Central Processing Unit?

○ **Central processing unit:**

Central Processing Unit (CPU or processor) is a central part of a computer (and can

be referred to as the computer's "brain"). It manages all other computer parts, monitors their mutual communication and performs arithmetic-logical operations. Processor speed is measured in hertz (or megahertz or gigahertz). Most famous manufacturers for personal computer processors are Intel and AMD.

25. Define Read Only Memory (ROM) and Random Access Memory (RAM)?

- **ROM:**

ROM (Read Only Memory) is a type of permanent, internal memory that is used solely for reading. All BIOS programs are stored in ROM.

- **RAM:**

RAM (Random Access Memory) is a working memory in which analyzed data and programs are stored, while computer runs. It allows reading and writing data, and it is deleted when the computer shuts down.

26. What is Technological innovation and difference in learning?

This story leads us to the definitive question which underlines the entire course: how can we, with the appropriate use of technological innovation, cater to people who are different in education? By different we refer to people who cannot adapt to a standardized educational system, despite possessing innate talent. By answering this question gradually we will be able to achieve the objectives of this school.

There are two types of individual in this world. One who can adapt themselves to societal norms and as such are rewarded with material success. Then there are those who are unreasonable and unable to fit in. According to George Bernard Shaw all the progress in society is based on these people. Based on a set of limited resources and ever growing users, only with innovation will we be able to distribute these goods more evenly rather than letting their accumulation confined to the few. Our current schooling system too is standardized and has hardly evolved for ages.

Instead of the person having to adapt to the system, the system should be flexible enough to adapt to the individual user. As mentioned earlier, at that time, the number of matriculation scholarships was a criterion of academic distinction. In our school there were about three hundred students who would appear for this examination. Of these who had adapted themselves well to the system would, distortion, a numbering about fifty or so, would be sponsored by the school in terms of fees? Another advantage would be that this group would be groomed by the best faculty. This may seem a little harsh but what choice did the school have with three hindered students and limited resources? If the limited faculty members were distributed for the entire student body the marks of all would have improved marginally, but there would be no scholarship students.

This no longer holds true! Technology allows us to address this and other educational issues. Of this elite group of students fifteen to twenty students were identified as sure in

getting a scholarship and were further segregated. These students, of which I was one, had become versed in meta-cognition: they were able to learn on their own.

Consequently, they were treated differently and enjoyed a great deal of independence, i.e. if they wanted to study on their own. Of the remaining two hundred and fifty they had to pay and make their own way. The talented students I had mentioned earlier were nowhere on the scene. It was surprising that a mediocre student like me could through hard work reach a position where these more talented students could not? In the sixth standard, those who could not pass would not be detained. A standard practice in private schools at that time, i.e. they would still be promoted to class seven. There was no feedback and parents would often wonder why despite failing a grade the child was still promoted? From the perspective of school they could not be detained since there were three hundred students coming from lower classes and there was not enough space to retain these failed students.

Apart from our school there was government pilot school in the same vicinity; a good school with young qualified faculty. It was a practice in that school that any failed students would not be retained. Still there was no feedback mechanism, in both type of institutions there was no feedback as to how a student was performing and what help he required. In one you would be promoted despite failing while in the other you would be dropped from the school for failing. Things have not changed much; there is still no feedback apart from annual exams and various tests in between. With technology the equation has changed, we can now instantaneously gauge a child's academic level and requirements: we can see clearly where, how and why a student went wrong in a particular subject.

27. Define BIOS?

BIOS (Basic Input/output System), can be defined as a program which is located in a separate ROM on the motherboard, and defines, as the name suggests, basic input/output system, is a good example.

28. Define Sound Card?

Soundcard is a device used for sound creation and production by means of computer speakers.

29. Define CD-ROM and DVD Drive?

- CD-ROM drive is used for reading CD media.

- DVD drive is used for reading DVD discs. DVD disc capacity ranges from 4.7 to 18GB

30. Define Graphic Cards?

Graphics card is responsible for image processing and displaying it on a monitor. It has its own graphics processor and memory. Image quality depends on the strength of these components.

31. Define Modem and Connectors?

- **Modem:**
 - Modem enables computers to communicate via telephone lines. They connect computers to the Internet.
- **Connectors:**
 - Connectors or ports are slots visible in the back and the front side of a computer.

32. Define Universal Serial Bus (USB)?

Universal Serial Bus (USB) is used to connect various devices (mouse, keyboard, USB memory).

33. Define Serial Port and Parallel Port?

- **Serial port:**
 - Serial port is used for example in connecting a mouse (labeled COM1 or COM2).
- **Parallel port:**
 - Parallel port is used for connecting a local printer (LPT1 or LPT2).

34. Why is the earth orbiting the sun?

I remember in our Physics class at UET, first year, a very intelligent and naughty student once addressing the Physics professor and enquiring if he could demonstrate a lab

experiment which would rationally demonstrate the fact that Earth spins on its axis and that the sun is not moving? The professor said we should go next door to the National Science and Technology Museum and see for ourselves; since there is an experiment which demonstrates this. We were all very excited and a time was set for next week. At the museum we were very warmly received and shown a huge pendulum which was hung with a very long length of wire. The pendulum was driven by a motor and remained in perpetual motion. Naturally, a motor was required since otherwise it would stop due to friction. On the sides of the pendulum there were sticks, which upon collision would fall down. It would do so one side and then the other with a slight shift; this shift demonstrates the circular motion of the earth. This was stated with such authority that no student with the exception of one questioned the museum representative's assertion. He was told that he was at the back and how he could understand. Now the dilemma for most students was connecting the shift in the to and fro motion of the pendulum to the spin of earth. The expensive demonstration was quite futile. A better option would have been to carry out a thought experiment, which would also entail some absorption activity; much like Galileo did several hundred years back.

35. Define Network Port and Firmware?

- **Network Port:**

Network port is used for connecting computers to a network.

- **Firmware:**

Firmware is used for connecting computers and audio-video devices (digital cameras, etc.).

36. Define Measurement Units?

Bit (binary digit) is the basic unit used to measure the amount of information. A byte or octet contains eight bits.

- 1 KB (kilobyte)- 1024 B (approx. 1000 B)
- 1 MB (megabyte)- 1024 KB (approx. 1000 KB)
- 1 GB (gigabyte) -- 1024 MB (approx. 1000 MB)
- 1 TB (terabyte) – 1024 GB (approx. 1000 GB)

37. What are the Limitations of ICT?

- **Limitations of ICT:**

- ICT supposedly endangers the child's social development
- ICT is supposedly to endanger the child's intellectual development.
- Decreasing of child's emotional development
- Child's health is endangered.
- Misuse of ICT A
- Complete control over the child's usage of ICT is today
- Practically impossible
- Expensive

38. Define Mouse?

○ **Mouse:**

Mouse is an input device that facilitates work with the graphical user interface (GUI). The mouse transmits hand movements and the screen displays the cursor (mouse pointer) movements. They are divided into mechanical and optical (with respect to a transfer movement), and wired and wireless (with respect to connection).

39. Define Scanner?

○ **Scanner:**

Scanner is used to load data (image, text, etc.) from the printed material into a computer. The result of scanning is an image, but with special programs, if we scan the text, we can get a text as a result. Software used to recognize text from image is called a text recognition tool.

40. Define Printer?

○ **Printer:**

Printer is a device used for printing data from a computer onto a paper. We distinguish between local printer (connected directly to the computer) and network printer (connected directly to network using a network card). Also, printers also differ according to print technology: dot matrix, laser, inkjet, thermal printer and plotter.

41. Define Dot Printer and Laser Printer?

○ **Dot Printer:**

Dot matrix printers are the oldest, with the lowest price of print per paper, they are slow, make a lot of noise while printing, and are mostly appropriate for printing text.

○ **Laser Printer:**

Laser printers are similar to photocopy devices when it comes to technology. They have exceptional print quality, speed and are quiet. Downsides of laser printers are their high price and high price of toners.

42.Describe merits and demerits of HTML?

○ **Merits of HTML:**

- HTML is simple but powerful formatting language to use.
- The simplicity allows anyone to create web pages.
- HTML documents are device independent (works on any platform)
- The web pages can be linked together using links
- The pages can be updated at any time.

○ **Demerits of HTML:**

- HTML does not have any programming capabilities.
- Require expertise knowledge.
- Time consuming.
- Difficult to remember coding

43.Briefly describe Quality control and assurance?

We were discussing that in car pipeline, inspection couldn't be done after 10 steps as we do in education after 10 years assessment. This happens in every step in pipeline so that there wouldn't be any compromise in the car quality but we do compromise in education and we do assessment in the end of the year and still don't learn anything from the result and promote them in higher classes, this a blunder we are doing. Alright, what happened next? He went to the Toyota company after coming out of Ford Company and said I'll set the front the seat here as well, although I have learnt from that factory about seat fixing but you also tell me about the steps verbally. Then he got surprised there, they told him to be master in 1st step first and then we will tell you the second step before that we will not tell you the second step. This was the basic difference and what happens because of this? This change and give impact on the economy of countries; progress and demolishment.

Toyota defeated all the companies of America because of these reasons. These are the minute decisions which need to be taken. But if we will say whoever wants to be educated and he will be. If he doesn't learn so it okay then. They told him take one hour or six hour or one day to master the 1st step. But till the time you don't show us the full command on 1st step we will not tell you the second step. Learning time is variable but outcome is fixed.

Very basic thing is this audience; this is the purpose of this course. Summing up it in one line we can say that we should make the learning time variable and fix the outcome. Student should get time in all periods to learn and to make mistakes, give them chance and testing and assessment is essential. This is the lesson which we have learnt, if you do that you will see the difference, difference in cars and difference in the pipeline of education. Stew fixed all his steps brilliantly and he is the same Stew who was unable to fix the seat in previous company. He was unable to fix seat in time and because of they stopped the pipeline and here lies the secret of excel of Toyota in company. You can clearly see more Honda, Toyota and Corollas rather than Ford. This is the major reason. Inspection is must on every step otherwise you can see the consequence. Now, I'm quoting Claton who says what should be our future? The result of testing and inspection, in education its assessment and in factory it's inspection. Testing and inspection at every step is essential. Assessment or inspection is interpedently woven into content delivery with fixed results. This should be our aim in the course of education by using technology. This minimizes the repair work in the end of pipeline and there were essentially no dropout after 6th class or 10th class, which these students are not able to study anymore. That is geostrophic for school students that are geostrophic for motors. Technology and software provides the feedback loops to improve themselves for different kinds of learners. There is another blunder that we try to teach everyone with same method, Learning styles come here. So audience summing it up doesn't separate assessment and learning. Assessment helps learning and learning helps assessment that we have separated and experiments, we have separated completely. Experiments, labs of experiments, assessment and learning should be together like this. After that we will be able to produce scientist and they will be able to do discoveries.

44. Define Operating System?

- **Operating System:**

Operating system is a program which manages computer hardware. First computers did not have operating systems; they had programs that were directly loaded into the computer (e.g.

punch cards). Today, computers have an operating system which loads into the computer's

memory during its startup. Computer functions are based on its operating system.

Within

operating system, drivers (responsible for the functioning of a computer) and various utility

programs (responsible for the functionality of a computer) are installed. The most famous

operating systems are:

1. Linux (Debian, Ubuntu, Fedora, Knoppix) open source software
2. Microsoft Windows (XP, Vista, 7) proprietary software
3. Mac OS X (Cheetah, Panther, Snow Leopard) proprietary software

45. Define Application Software?

- **Application Software:**

Application Software (Utility programs) are all programs that users use to perform different tasks or for problem solving. Users, according to his/her needs, install the appropriate utility software. Computer functions and tasks that computers can perform are defined by the installed utility software. Utility software can often cost more than computer hardware unless the software is open source.

46. What is Importance of Audio Visual Aids?

○ **Importance of Audio Visual Aids:**

- They are best motivators
- Reduce verbalism
- Gives clear images
- Vicarious experience
- Provide variety in methods of teaching and learning
- Freedom to child
- Opportunities to handle and manipulate
- Contributes to increased retentively
- Based on maxims of teaching
- Helpful in attracting attention
- Helpful in fixing up new learning
- Saving energy and time
- Gives reality and vividness to learning situations
- Meeting individual differences
- Encouragement to healthy classroom interaction
- Spread of education on a mass scale

47. Define Computer Network?

○ **Computer Network:**

Computer network is comprised of at least two, connected, by wire or wireless, computers that can exchange data i.e. communicate. There are many reasons for connecting computers into a network, and some of them are:

- Exchange of data between users that have network access,
- Access to shared devices, such as network printers, network disks, etc.,
- Enables user communication and socializing, etc.

LAN (Local Area Network), WLAN (Wireless Local Area Network and WAN (Wide Area Network)

LAN (Local Area Network):

It is defined as, a network that covers a relatively small geographical area- it connects computers within a firm or household by wire.

WLAN (Wireless Local Area Network):

It is defined as, a network that covers a relatively small geographical area - it connects computers within a firm or household wirelessly.

WAN (Wide Area Network):

It is defined as, a network that covers a relatively large geographical area - it connects a greater number of computers and local networks.

48. Define Internet, Intranet and Extranet?

- **Internet:**

Internet (network of all networks) is a global system comprised of interconnected computers and computer networks, which communicate by means of using TCP/IP protocols. Although, in its beginnings, it emerged from the need for simple data exchange, today it affects all domains of society.

- **Intranet:**

Intranet is a private network of an organization to which only authorized employees have access (login and password).

- **Extranet:**

Extranet is part of Intranet, to which independent collaborators have access.

49. Different Connection Methods of Internet Mobile?

Internet is connected using a mobile network (GPRS, EDGE, UMTS, HSPA)

- **Satellite:**

These are commonly used in parts of the world where there is no proper infrastructure and there is no other way of accessing the Internet

- **Wireless (Wi-Fi):**

Data is transferred between computers by using radio frequencies (2.4 GHz) and the corresponding antennas.

- **Cable Internet:**

It is used in connecting to the Internet through television cable network using a cable modem

- **Broadband:**

It is characterized by a high-speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your personal computer. In the beginning of broadband Internet access, due to underdeveloped communication infrastructure, Internet providers charged based on the data traffic but not time spent on the Internet (unlike dial-up Internet access). Today, in large cities, telecommunications infrastructure is developed, therefore Internet providers do not charge money based on the time spent on the Internet or the amount of transferred data but they do charge by access speed.

50. Define E-Commerce and E-Banking?

○ E-commerce:

E-commerce is a form of trade that allows customers to browse and purchase products online.

○ E-banking:

E-banking allows users to have control over their accounts (view balances and transactions), transactions from one account to another, credit payment, shopping vouchers for mobile phones etc. The benefits are saving time (which would otherwise be spent waiting in lines), lower service fees, and access from anywhere, anytime.

51. Define Episcopes & Epidiascopes?

- Episcopes are used for the projection of opaque objects.
- Pictures, photographs, drawings or an opaque material within the size of the platform of the episcopes can be projected using Episcopes.
- Episcopes are used to enlarge book diagrams and illustrate complex pictures.
- During Epi-projection the class room should be totally darkened in order to get a clear and bright image.
- Epidiascopes can project opaque as well as transparent objects.

52. Describe DLP Projector (Digital Light processing Projector)?

- Most advanced and sophisticated projected aid.
- Brighter image
- Lightest and smallest

53. Define E-Learning and advantages of E-learning?

○ E-learning:

E-learning consists of all forms of learning and/or knowledge transfer that are based on electronic technologies. This term will mostly be used to describe learning and/or dissemination of knowledge without direct teacher-student contact, while using ICT technology.

○ Advantages of E-Learning:

Advantages of e-learning greatly coincide with learning from books, with a difference that the books come only in text and graphic form on paper, while e-learning materials have multimedia features, and the ability to remotely monitor lectures in real time. To illustrate, physicians can observe medical procedures that are currently taking place in another hospital; students can attend classes that are currently taking place at another university.

Some of the advantages for students:

- Adaptive time, place and pace of learning
- Access and availability of education
- Unlimited repetition of lectures
- Reduced cost of education
- Multimedia environment (video, audio, text)

54. Describe Benefits of E-Learning for Teachers and Institutions?

Some of the benefits for teachers and educational institutions:

- Reduced costs of conducting teaching (reduced travel costs and accommodation, reduced or eliminated costs for classrooms, etc.).
- Much larger education user base
- Once created, e-education allows educational staff to invest a much shorter time commitment to implement the respective education.
- Simple and easy upgrade of materials, etc.
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55. Define IM and VOIP?

○ IM (instant Messenger):

IM (Instant Messaging) service is used for instant message exchange and real time communication. Besides textual communication and video conferencing, they allow file transfer as well. Some of the popular IM services are: Windows Live Messenger, Skype, and Google Talk.

○ VoIP (Voice over Internet Protocol):

VoIP (Voice over Internet Protocol) is a term for technology that enables digitalization and transfer of sound (and multimedia sessions) over the Internet. Therefore, it enables sound communication much like the telephone call. In short, VoIP protocol enables audio/video communication between users

56. Define Blog (Web Log)?

Blog (web log) is a form of Internet media in which users can re-post or write their opinion on any topic. It can be personal (a form of Internet diary), thematic, or in the form of journals.

Blogs are popular because end-users do not need to have advanced computer knowledge, or knowledge of webpage creation, in order to be present on the Internet and write about various topics, i.e. state their opinion. Blogging has greatly facilitated the exchange of interests, ideas, opinions, and contributed to realization of human right to free expression. Some use blogs to share their art, personal aesthetic, or to collect virtual clippings from websites online.

57. Describe Social Networks and Forum?

- **Social Networks:**

These are free online services that provide users with various forms of presentation and communication. Some of the more popular social networks are: MySpace, Facebook, Twitter, and Google+.

- **Forum:**

It is defined as a service that allows users to exchange opinions. It can be compared to bulletin boards on which participants leave their messages, while others leave their comments.

Examples of forums are: Reddit, TED Conversations.

58. Define Computer Virus?

Virus is a malicious program with the possibility of self-replication. In a computer it seeks other files, trying to infect them and the ultimate goal is to spread to other computers.

Depending on what malicious code it contains, it may cause more or less damage to your computer.

59. Define Trojan Horses?

Trojan horses can be presented in the form of games or files in e-mail attachments, with different purposes, such as; to allow remote control to your computer, delete data, allow spreading of viruses, send spam from the infected computer, etc. Unlike viruses and worms, it does not replicate and it needs to be executed, that is, it is not autonomous.

60. Define Worms?

Worms like viruses, they possess the ability of self-replication, and however they cannot infect other programs. Their usual intent is to take control of your computer and allow remote control by opening, the so-called, and 'backdoor'

61. Define Web 2.0?

Web 2.0: Contrary to what many people think, Web 2.0 is not a new version of the World Wide Web. The term arose as the name of a series of conferences, the first of which was held in 2004: <http://www.web2summit.com>. Essentially, Web 2.0 is an attempt to redefine what the Web is all about and how it is used, for example new Web-Based communities using Blogs, Podcasts, Wikis and Social Networking websites that promote collaboration and sharing between users - in other words, a more democratic approach to the use of the Web.

62. Define VGA?

VGA is abbreviation for Video Graphics Adaptor. An older type of Video Card or circuit board used to control the output on a computer Display Screen. VGA cards were superseded by SVGA cards. See Section 1.1.1.4, Module 1.2, under the heading Graphics card.

63. Briefly Disruptive innovation?

It is an inherent part of the Capitalist system; that based on technology that an improvement in quality, reduction in size, consequently price and improvement in features occurs, leading to disruption in existing offering – replacing businesses which are no longer sustainable. As discussed earlier Sony's Walkman replaced the large transistor based systems of RCA and Zenith. Similarly, Sony's walkmans were displaced by Apples iPod: a product which was more reliable and more efficient despite being slightly expensive.

Now we turn towards computers, a more relevant example since an integral part of our course. We had already discussed the IBM 11 30, which came in 1965 and was priced at US \$32,000; its memory was approximately 64k. While I was studying the machine was available at our college. In those days only universities and other large institutions could afford these machines. We had also discussed that computer technology can be addictive since they provide feedback without embarrassment. How were then these replaced or rather disrupted? IBM still is one the largest global corporations which happened due to a lot of flexibility and maneuvering on their part. A fact, which can be illustrated by a change in their corporate culture, ten years ago ninety percent of their employees came to work and punched in their cards and on leaving punched out, to determine if they were complying with office hours. If they did not comply their salaries were deducted. These days seventy or so percent of their employees work from home. IBM earlier competed with Data General and Digital Corporation DEC. IBM was focused on business solutions while Data General and DEC were focused on educational institutions. A person doing his PhD from the States in the 80s invariably used a DEC VAX, 730 750, etc. Turning to a book, a friend had gifted me with a novel, 'The soul of a new machine' by Tracy Kidder about thirty years ago. The novel deals with the competition between Data General and DEC. DEC's 780 entered the market earlier. Data General wanted to understand what innovations were associated with the machine, prior to launching their computer.

64. Describe Prior knowledge as the foundation?

We welcomed you in new module. I hope you are enjoying this course ICT in new education. I have told this quotation many time, in future I will tell you some one. May be you remembered the name. It was quotation from very famous learn Science scientist. "Most important in learning is your prior knowledge" Prior knowledge is the foundation of the building. If the foundation is weak, we cannot construct a building properly. We have to check our prior knowledge, is it correct or not, it is on certain level and technology help us. This slide you saw many times, and i show again because we will do some real example.

I am telling you stories and philosophy till; we converted into picture, so you can view it through different angles. In this slide we are showing technology, pedagogy and content. We can never separate them. We have to combine them and strange thing happened after combination. Children try to discover and enjoy be self. Children become excited and intrinsic motivation. Then a strange play started after little effort. You can do wonders. The second most import thing in learning how does you integrate new knowledge with prior knowledge. The first test technology does. Is your prior knowledge appropriate or not.

The second test is how you integrate new knowledge with prior knowledge. I repeat this thing may time. We have tools for pedagogy, which are ABSORB--DO connect. This thing comes everywhere. In another book this tool is written in another way, select--organize--integrate. Prior knowledge is most important, where we start it. Second most important thing how do you integrate it. Technology tries to help you on every stage and have you where something happens not correct. Our all student are intelligent and sharp as compare to other nations.

If we learn to use mixture of pedagogy and technology, they do wonder. Responsibility is ours. We start from discovery base learning. One person came with prior knowledge and he have to reach the top of hill which his final destination. In term of learning, there are so many paths. This is our work to find which path is most suitable for which children.

Different children will choose different way pedagogy and technology help them to reach there. How much you do struggle with excitement and enthusiasm. The combination of technology, pedagogy and content help them to achieve their goals. This programmer can be seen by medical, science, pre engineering, arts and different discipline of children.

You can find the example of every type , some are close to physics, math , English language and some are simple. Today we selected the example of "finding the area of something" What is the area? Before I tell you what is the formula of area? How do you calculate it? First we know that what the area actually is. Some picture does not construct in your mind about the area. How many book do you read and memorize the formula of area, this is not deep learning. When I pronounce the word area, what is the thing come to

your mind? Would you please think about it for two minutes? We will start this thing from the next module.

65. Describe Client and clipboard?

- **Client:**

Client is referred to as a computer that receives services from another computer. A standalone computer on your desk which you use to browse the Web is a client, and the computers from which World Wide Web files are downloaded to your computer are servers.

Similarly, a computer (also known as a workstation) connected to a Local Area Network (LAN) is a client that can receive information from and send information to the server that controls the LAN.

- **Clipboard:**

Clipboard is a temporary storage area in a computer's memory. It may be used, for example, to store text that you are in the process of copying and pasting from one section of a word processed document to another section in the same document or to another document. You should find a clipboard viewer program on your computer, which enables you to see what is currently being temporarily stored in the clipboard.

66. Difference between Content Management System and Learning Management System?

Both these applications have important place in an organization. The differences between content management systems as compared to learning management system are significant in their core functionality. For students or learners, a CMS is more of a passive application where they can simply view documents. With an LMS, learners have the ability to view, listen, and interact with the data. They have the options to try examples, complete assessments, and give evaluation or feedback on courses.

67. Define Compatibility?

Compatibility is referred to pieces of hardware and/or software which is capable of being used together is described as compatible.

68. Describe different types of Charts and their Usage in Classroom Charts?

○ **Chart:**

- A chart may be defined as a combination of graphics and pictorial media designed for the orderly and logical visualizing of relationship between key fact and ideas

○ **Different types of charts are:**

- Display chart
- Flow chart
- Tabular chart
- Tree chart
- Time chart
- Flip chart

Display/Pictorial Chart

The subject matter is illustrated through pictures, diagrams, graphs, sketches and language.

This is the most commonly used type of chart in classroom teaching

Flow/process chart

- Flow charts are also known as organizational charts.
- They usually represent a process or relationship.
- Functional relationships in such charts are usually shown through various symbols and forms like rectangles, circles, lines, arrows and colors.

Tabular Chart

In table charts the data is presented in a tabular form.

- It can provide a quick comparison and contrast among various ideas and events.

Tree Chart

- The figures drawn in these charts resemble a tree.
- Generally used in showing development, growth

Time Chart

- Time charts are used to indicate the time sequence of a series of events.
- Arrange all the happenings important to the development of a process, organization or nation in a chronological order.

Flip Chart

- It consists of a pad of large paper sheets fixed to the upper edge of a board.
- Commonly used for presentations.

69. Define Incorporating technology and pedagogy?

Moving to a more complex shape with eight sides, an octagonal, we need to see how area can be calculated. There is no need of a formula rather by fitting the square in the octagon we see that the area left consists of triangles. We know how to calculate areas for both shapes and consequently work out the area for an octagonal. One step leads to another; standing at the bottom of a mountain it may seem unattainable yet the journey begins with one step. On the way one enjoys the various sites and challenges, which

become part of his experience. We need learning to be in this form where it becomes deeply etched into a student's mind. Now how do we calculate the area of a star? Can the squares be superimposed? We need to actually do this and forms a critical part of the learning process, i.e. trial and error. The making of shapes has always fascinated children and by teaching them in this way it becomes a game for them.

70.What are Different Types of Boards and Their Usage in Classroom?

○ Display Boards:

- Display categorized into three main types according to the function to the functions they perform.
- Motivational Display: Display that attracts and appeal pupils and stimulates their curiosity is called motivational displays.
- Developmental Displays: Displays that ass information and ensure active involvement of pupils in the development of lesson is called developmental displays.
- Summery Display: Displays that are used for review and consolidation of a lesson are called summary display.

○ Black Board:

- Most basic, widely used tool for display
- Teaching cannot be successful without proper use of a Black Board.
- Black Board planning is a precious skill to be developed by a teacher.

○ Roll up Board:

- It is that which can be conveniently rolled up.
- These are usually made of thick resin cloth.
- The charts and pictures can be drawn on it advance and presents as and when required.

○ Peg Board:

- To display flat and three dimensional materials
- Use hooks and clips to display materials.

○ Hook and Loop Board:

- Intended to suspend heavy 3D objects as well as flat materials.
- Objects can be easily fixed and removed.

○ Flannel Board:

- Flannel cloth used to make this board

○ Magnetic Board:

- A sheet of iron that attracts a piece of magnet can be used for magnetic board.

○ Plastic graph Board:

- Any smooth polished surface like rigid plastic sheet and serves as plastigraph board.

○ Bulletin Board:

- Used to display bulletin, News items, announcement etc.

○ **Marker Board:**

- Large plastic board suitable for writing or drawing with markers.
- This can be used the same way as chalk boards.
- It can be sometimes used as a projection screen also.

71. Define Islamic geometrical patterns and architecture?

The arches are Islamic patterns in three dimensions; a sight which bewilders the imagination. On observing them the first time one was faced with mixed emotions; happiness that this wonders belonged to our heritage and sadness on what was lost. Turning towards analysis we state a problem, a problem by the craftsmen of those days who had no computers, calculators or slide rules to help them. In those days two under root could not be worked out. Can you draw square of area two or given the square of area x , can you draw a circle of area $2x$? Do we just the initial area? That would be wrong since we would get four times the area. Looking at this visually this gives the clearest idea. Visual proof is a key methodology in the science of pedagogy. What we are trying to do is to establish a pedagogical methodology in arriving at solutions/ proofs which are not solely confined to the current area problem which we are dealing with. Even if provided with the appropriate methodology we should understand the why the method works— how does it work?

○ **Encryption:**

Encryption is a system of coding that helps prevent access to private information on computer networks or on the Web.

○ **Feedback:**

Feedback is an automatic response from a computer, which may take the form of text, image, audio, video or any combination of these, to a learner's input. Input from the learner may take various forms, e.g. (i) clicking with the mouse to select an answer in a multiple choice exercise, (ii) typing an answer at the computer keyboard, or (iii) speaking an answer into a microphone. Feedback in interactive language learning materials should go beyond a "boing" (wrong) or "applause" (right) or "try again" message and attempt to mimic the "live" situation when using the language results in either a response from the other person or an action showing that the language used was appropriate. Feedback is often described as intrinsic (implicit) or extrinsic (explicit).

72. Use of Three Dimensional Aids in Classroom

Three Dimensional Aids

○ **Models:**

- Models are concrete representations of objects.
- Size should be suitable
- There are two types of model
 - a) Working model
 - b) Static (still) model

c) Working model is more effective than static model **Objects**

- Objects may be defined as the real things from their natural settings. Eg. Rock, Coin, stamp etc. **Specimen**

- A specimen may be defined as a typical objects or part of an object which has been removed for convenient observation.

- It may be a representative of a class or group of similar objects.

○ **Mockups:**

- A mockup is an operating model usually of a process, designed to be worked with directly by the learner for specific training or analysis.

- When direct firsthand experience is either impractical or impossible, mockup can be used.

- Mockups are an imitation of the real process. E.g. Mock Parliament **Diorama**

- A diorama is a three dimensional scene in depth, incorporating a group of modeled objects and figures in a natural settings.

- A diorama scene is set up on a small stage with a group of modeled objects that are kept on the foreground and blended into a painted realistic background.

- For representation of depth, object kept at the background are made considerably smaller to create illusion.

○ **Puppets:**

- Different types of puppets are used to make the learning object more realistic.

- Most commonly used puppets are hand puppets, glove puppets, finger puppets, rode puppets, string puppets, and shadow puppets etc. **Globe**

- A globe is a three dimensional scale model of Earth or other celestial body such as a planet.

- The term globe is used only for models of objects that are approximately spherical.

73.Describe IP address?

IP Address is short abbreviation for Internet Protocol Address. The unique numerical address of a computer on the Internet, expressed as four sets of numbers (maximum 3 digits each) separated by dots: e.g. 150.237.176.24 for one of the computers at the University of Hull - where the ICT4LT website is located. Computers on the Internet are nearly always referred to by more memorable domain names, which are mapped onto their IP address by special Internet computer known as name servers.

74.What are the problems in the use of Audio Visual Aids?

○ **Problems in the use of Audio Visual Aids:**

- Apathy of the teachers
- Ineffectiveness of the aids
- Financial restrictions

- Absence of electricity/infrastructure
- Lack of facilities for training
- Activity Aids

75. Define Learning Framework?

We begin this module with two great names. Charlie Chaplin, a name you are all familiar with. Your elders would be even more familiar. Using silent movies as his medium he was able to explain what he wanted to convey through gestures and his actions; he was able to achieve phenomenal popularity. How did he manage this?

The other great name, which everyone is familiar with, is that of Einstein. Here we will not discuss what he said about education and technology; this will be dealt with later. What I want to narrate is his meeting with Charlie Chaplin, in which Einstein had this to say:

“What I admire most about your art is its universality. You do not say a word and yet the world understands you.”

From a learning, technological, multimedia and pedagogical perspective this ability, to be able to explain what you are trying to convey, is critical. In reply to “Einstein’s comments Charlie Chaplin said, “It is true, but your fame is even greater. The world admires you, when nobody understands what you say.” Now this could be taken as a bit of levity of Charlie Chaplin’s part or a serious comment, since at that time technology was not sufficiently advanced to allow verification of Einstein’s theories. Even through mathematics -a lesson we try and will try to convey in all mathematical teaching, that actual interpretation of equations and their result to actuality may differ -Einstein reached some conclusions which even he considered erroneous. The x in an equation has to mean something and how it relates to us; on its own it has no relevance. For instance one of Einstein’s discoveries of phenomena with infinite mass and negligible volume, where everything was being pulled towards it, even light; was something which he could not intuitively comprehend. This link/relation is critical and unfortunately for the most part missing in mathematical pedagogy and to some extent in physics. We do not develop the required intuitive correlation. Having rejected his own findings, it was later discovered by scientist, through modern technology, that Einstein’s prediction about Black Holes was quite correct.

Now turning to this course, it is about; Mathematics, Physics, Chemistry, but more importantly the ability to understand and the ability to convey through the use of technology to make it more effective. Utilizing technology we can perhaps comprehend things which earlier eluded us, for example in the instance of black holes. Einstein also said that due to gravity space gets curved. What does this imply mathematically, what are its equations and ramifications?

When Albert Einstein met Charlie Chaplin, Einstein said, “What I admire most about your art is its universality. You do not say a word, and yet the world understands you!”

It's true, replied Chaplin. But your fame is even greater; the world admires you, when nobody understands what you say!

ABI/Inform

It is a database of articles in more than 800 business-related periodicals and specialty publications since 1986.

Access Provider

It may be defined as, the company that provides Internet access and, in some cases, an online account on their computer system.

76. Define Constructivist conversations?

In this module we discuss how Piaget interacted with children. The important point is that they should be dealt on their own level, with consideration and respect. He began by enquiring what makes the wind. This is not a standard text book question and one which children often ask. Now let us look at how children think. He put this question to a five year old girl. Her answer was the trees. Piaget did not rebuke her, rather he engaged her in further conversation to understand her reasoning – the why. Julia replied, I saw them waving their arms. Now this is an interesting concept are the tree waving due to the wind or their waving causing the wind? Any intelligent person can conceptually confuse the two and we cannot say that Julia's answer was wrong. The phenomenon entails an observation based on a particular perspective - what is the cause and the effect. Piaget enquired further, how does their waving their arms cause wind? Julia waved her hands in front of Piaget's face and said this is how, but the trees are bigger. Naturally, Piaget felt wind on his face. Here an important point needs to be mentioned; that children learn from their experience and interpret information based upon this. Here through a small action on her part she has sought parallel in large event. This is termed as learning, despite the fact that the answer may be incorrect. Piaget got the girl to interpret, through conducting a small (her hand) experiment, a large phenomenon – a brilliant achievement. We can achieve this through technology.

Based on this he moved on to the question that what makes the wind on the ocean. By eliminating the presence of tree, Piaget was trying to create cognitive disequilibrium in the girls mind. "Piaget's concept of constructivism relates to his studies on how knowledge is internalized and how people learn. Humans according to Piaget internalize knowledge through experience and make sense of these experiments through adaptation of involving processes of assimilation, accommodation, equilibrium and disequilibrium. It is through these three processes that we learn, outgrow some ideas and adopt new ones. Now let us observe how the little girl attempts to outgrow her faulty thinking. The absence of trees causes her to contradict her earlier thinking. This dissonance and conflict in her mind is crucial to learning. An instructor for example seeks to stimulate conceptual change by challenging a student's existing concepts, in order to create cognitive

disequilibrium. The student will try to restore equilibrium and resolve a problem through a process of disequilibrium and re- equilibrium, the student construct new cognitive structures.

A/D Converter:

A device used to convert analog data to digital data. Analog data is continuously variable, while digital data contains discrete steps.

Additive Primaries:

It is red, green, and blue are the primary colors of light from which all other colors can be made.

77.Intersection of pedagogy, Content and Technology

In this module we look at how technology, content and pedagogy intersect: The triangulation of three contains a numerous hidden issues and what combination of expertise and multidisciplinary approach are required.

The appropriate person needs not be an expert in all three fields but should have working knowledge of each. The diagram presented here of constructivist technologies contains three circles; of technology, pedagogy, and content. At the intersection of three is where we want to create interactive learning applications. Currently, the most useful innovations, which most people would assume to in the field of a singular discipline, i.e. computer science; are in fact being done at the intersection of multiple disciplines.

Graduate students should need to remember a fact. For instance a computer science would not solely be asked to create program in C++, this program would be created to deal with a real life problem, requiring specific knowledge. I have enquired of many PhD's in education a simple mathematic problem concerning a circle, which is usually dealt with in the 6th or 7th standard. Their immediate reaction is to keep mathematics out since their field is pedagogy – it does not work this way.

Similarly, if the technology people start saying that they will not deal with pedagogy, in Physics Newton's laws, and just do programming in C++ - nothing will come out of it. Those universities imparting multidisciplinary knowledge and those individuals with these tools will be able to face the future successfully. Turning to content: suppose we have to design a content for a 9th or 10th standard book, an interactive book or for that matter a learning application or a learning game; the users while immersed in the experience should have their minute to minute progress recorded, their learning styles recorded. Naturally, there has to be content. It could be pure sciences or social sciences. From the social sciences geography is particularly suitable. One could transport the students to specific destinations, utilizing three dimensional animations, Salt mines in Khavera or Niagara Falls in Canada.

The next element is pedagogy and here we address this in reference to constructivist theory, which is currently the prevalent methodology. In this we have to understand how

people learn, epistemology, and how people do not learn and why do they not learn; the second part of the equation being of greater importance. Some people may have the prerequisite knowledge and intelligence yet have difficulty learning. Is their learning style different? The important point is that it is not their failure but rather ours. It is the fault of the pedagogical, technological content or the teacher.

78. Define ASCII?

It is abbreviation of American (National) Standard Code for Information Interchange. It is a standard character-to-number encoding widely used in the computer industry.

79. Define Binary and Binary file?

- **Binary:**

This is a numbering system with only two values: 0 (zero) and 1 (one).

- **Binary File:**

It is a file that contains more than plain text (i.e., photos, sounds, spreadsheet, etc.) In contrast to an ASCII file which only contains plain text.

Useful set of 'core' indicators for ICT

Given the great variety of circumstances and challenges, and the great variance in the use of ICTs in education from country to country, it would be unrealistic (and inappropriate!) to

attempt to formulate a uniform set of indicators that can be used to frame data collection for

ICT in education projects. That said, it would be quite useful to have a set of commonly agreed upon, 'core' indicators that can be used across countries. Important criteria to be observed in formulating these core indicators would include local relevance, reliability and

robustness when these are used for comparison of one ICT project or country with another.